NUMBER THEORY LADDER 2 OF 9

Prime factorisation, divisors and squares

Every number is a product of primes in one way only, and its divisors are the combinations of those primes.

1 / THE RULE AND THE PICTURE

Divisors are the products $2^i 3^j$, so $36 = 2^2 3^2$ has $3 \times 3 = 9$ of them

	3^0	3^1	3^2
2^0	1	3	9
2^1	2	6	18
2^2	4	12	36

d and $36/d$ sit opposite each other, and 6 pairs with itself

Writing $n = p_1^{e_1} p_2^{e_2} \cdots$, a divisor takes each exponent from 0 to e_i, so

$$d(n) = \prod (e_i + 1), \qquad \sigma(n) = \prod \frac{p_i^{e_i+1} - 1}{p_i - 1}$$

where d counts the divisors and σ adds them. For 36 we get $d = 3 \cdot 3 = 9$ and $\sigma = (1+2+4)(1+3+9) = 7 \cdot 13 = 91$.

2 / HCF AND LCM FROM EXPONENTS / 36 AND 90

The HCF takes the smaller exponent of each prime and the LCM the larger

number	2	3	5	value
$36 = 2^2 \cdot 3^2 \cdot 5^0$	2	2	0	36
$90 = 2^1 \cdot 3^2 \cdot 5^1$	1	2	1	90
HCF, the minimum	1	2	0	18
LCM, the maximum	2	2	1	180

The identity $\mathrm{HCF} \times \mathrm{LCM} = ab$ holds because $\min + \max =$ the sum of the two exponents, so $18 \times 180 = 3240 = 36 \times 90$.

3 / COUNTING DIVISORS / $d(n)$ AND $\sigma(n)$

The count is odd only when every exponent is even, that is when n is a square

n	factorisation	$d(n)$	$\sigma(n)$	note
12	$2^2 \cdot 3$	6	28	
28	$2^2 \cdot 7$	6	56	perfect
36	$2^2 \cdot 3^2$	9	91	square, odd d
60	$2^2 \cdot 3 \cdot 5$	12	168	
100	$2^2 \cdot 5^2$	9	217	square, odd d
360	$2^3 \cdot 3^2 \cdot 5$	24	1170	

A number with $\sigma(n) = 2n$ is perfect, and $28 = 2^2 \cdot 7$ has $\sigma = 7 \cdot 8 = 56$. Divisors pair as d and n/d, so they come in twos unless $d = n/d$, and that needs $n = d^2$.

4 / THE LOCKER PUZZLE / ONE HUNDRED DOORS

Locker k is toggled once per divisor of k, so it ends open exactly when k is a square

1	2	3	**4**	5	6	7	8	**9**	10
11	12	13	14	15	**16**	17	18	19	20
21	22	23	24	**25**	26	27	28	29	30
31	32	33	34	35	**36**	37	38	39	40
41	42	43	44	45	46	47	48	**49**	50
51	52	53	54	55	56	57	58	59	60
61	62	63	**64**	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	**100**

Students walk a corridor of 100 closed lockers and student s toggles every sth locker. Locker 12 is toggled by students $1, 2, 3, 4, 6, 12$, six times, and ends closed, while locker 36 is toggled nine times and ends open. The open lockers are the ten squares $1, 4, 9, \ldots, 100$.

5 / WHY FACTORISATION IS UNIQUE / EUCLID'S LEMMA

If a prime p divides ab then p divides a or b, and uniqueness follows

Suppose $p \mid ab$ and $p \nmid a$. Then $\mathrm{HCF}(p, a) = 1$, so Bézout gives $px + ay = 1$ for some integers, and multiplying by b gives $b = pbx + aby$, which p divides because $p \mid ab$. Cancelling primes one at a time from two factorisations of the same number then shows they agree. Among the even numbers alone the argument fails, since 6 and 18 cannot be split inside that set and $36 = 6 \cdot 6 = 2 \cdot 18$ factors in two ways.

ASIDE / HIGHLY COMPOSITE

360 has 24 divisors, more than any smaller number, which is why it rules circles and clocks.

LINKS TO OTHER SHEETS

The proof uses Bézout from *Euclid's algorithm and Bézout's identity*; the next sheet is *Arithmetic on a clock*.

Convention. $d(n)$ is the number of divisors, $\sigma(n)$ their sum; $\min$ and $\max$ are taken exponent by exponent.